Question 4: Install and configure Git on Ubuntu Linux, verify the installation, and set up global Git configuration (user name and email).

Marks: 5 | Course outcome: CO1 | Environment: Ubuntu 24.04 LTS (WSL2), Git 2.43.0, Jenkins 2.580.2 LTS

Git 2.43.0 and Jenkins 2.580.2 LTS were installed on Ubuntu 24.04.3 LTS (WSL2) under the user meenakshi. All commands were executed in the Ubuntu terminal.

1. System and Git installation

installation and verification

```
meenakshi@ubuntu:~$ lsb_release -d
Description:    Ubuntu 24.04.3 LTS
meenakshi@ubuntu:~$ sudo apt update && sudo apt install -y git
meenakshi@ubuntu:~$ git --version
git version 2.43.0
```

2. Global Git configuration

configuration commands

```
git config --global user.name "Meenakshi"
git config --global user.email "meenakshi16rp@gmail.com"
git config --global init.defaultBranch main
git config --global credential.helper store
git config --global core.editor nano
```

verification

```
meenakshi@ubuntu:~$ git config --global --list
user.name=Meenakshi
user.email=meenakshi16rp@gmail.com
init.defaultbranch=main
core.editor=nano
credential.helper=store
```

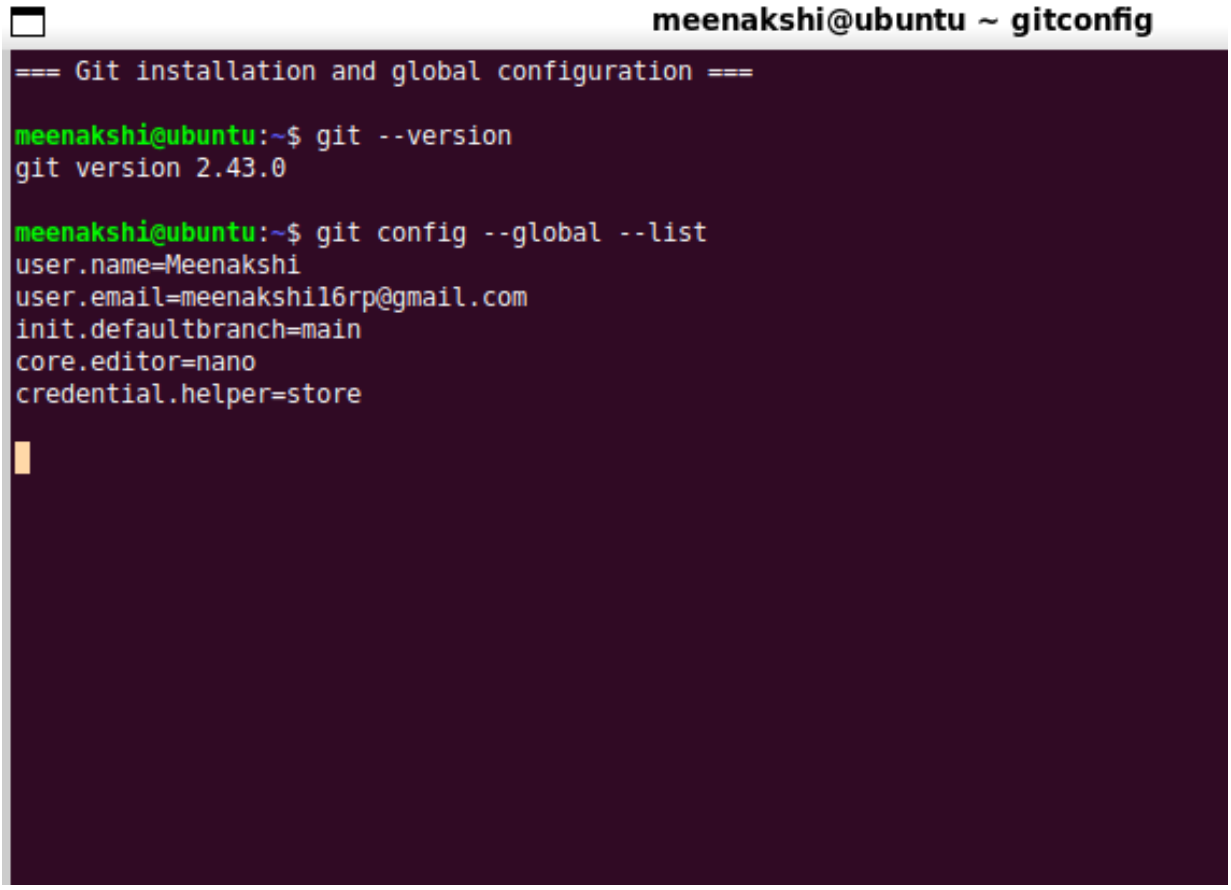


Figure: Ubuntu terminal: meenakshi@ubuntu prompt, Ubuntu 24.04.3, Git 2.43.0 and the global configuration

3. Jenkins installation

Jenkins LTS installation (systemd service on port 8080)

```
sudo apt install -y openjdk-21-jdk
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key -o \
      /usr/share/keyrings/jenkins-keyring.asc
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/" \
      | sudo tee /etc/apt/sources.list.d/jenkins.list
sudo apt update && sudo apt install -y jenkins
sudo systemctl enable --now jenkins
sudo systemctl is-active jenkins
active
jenkins --version
2.580.2
```

4. Jenkins first-run configuration

The setup wizard was completed in the browser at http://localhost:8080: unlock with the initial admin password, install the suggested plugins (Pipeline, Git, GitHub Branch Source, Credentials Binding, ...), and create the administrator account.

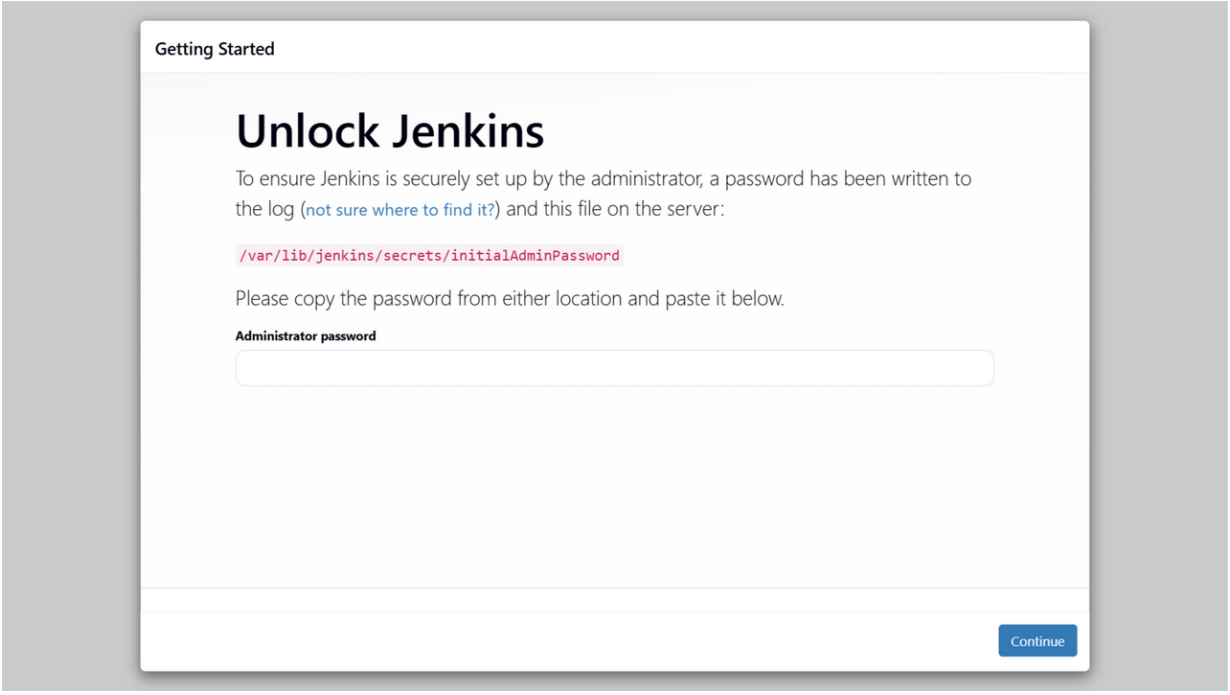


Figure: Unlock Jenkins - initial admin password from /var/lib/jenkins/secrets/initialAdminPassword

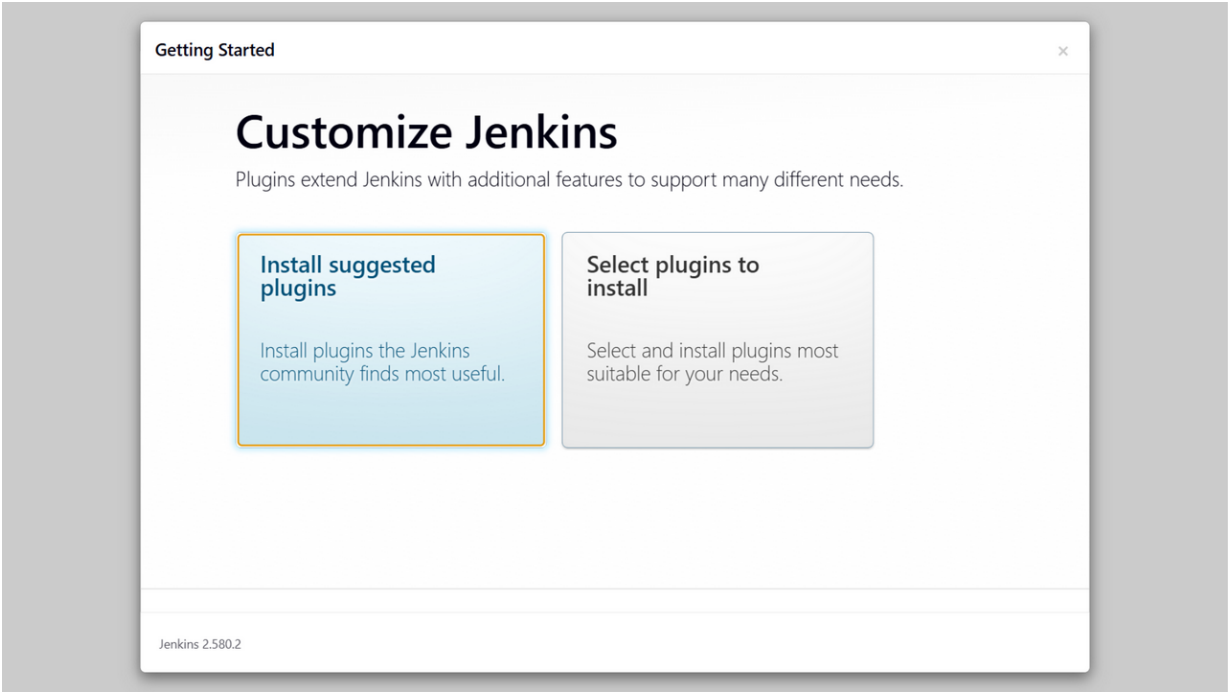


Figure: Customize Jenkins - install suggested plugins

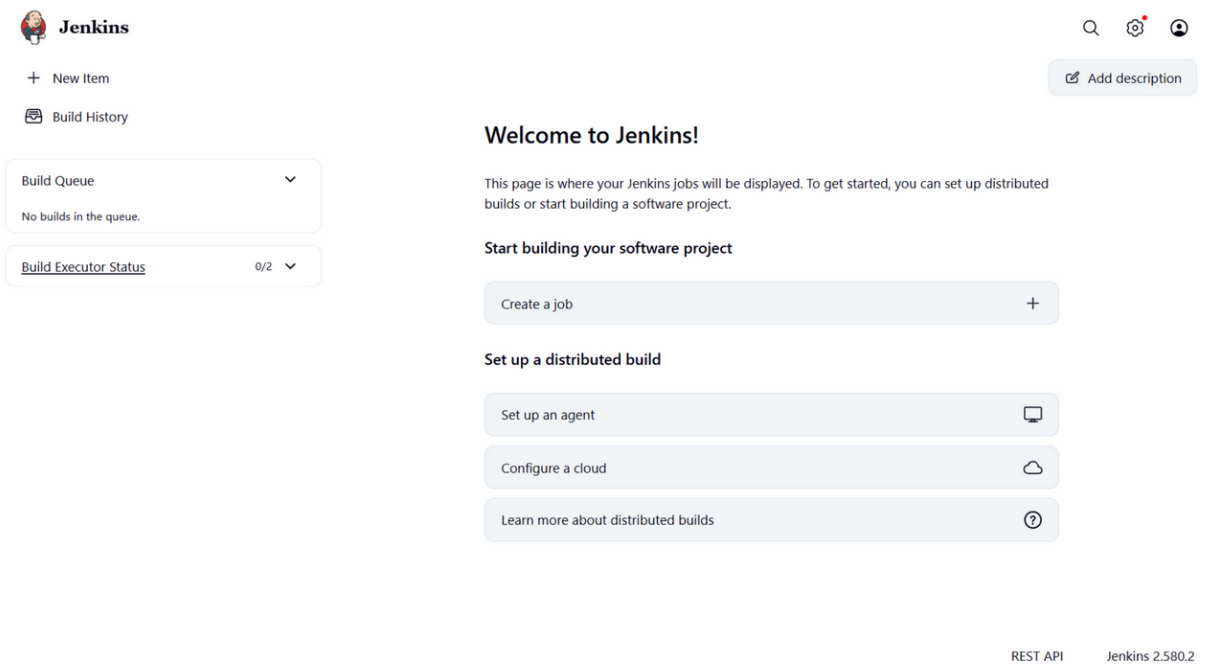


Figure: Jenkins dashboard ready (admin user: meenakshi)

5. Observations

- Jenkins runs as a systemd service (user jenkins, JAVA 21) and survived WSL restarts.
- The initial admin password file is removed after unlocking; plugins install directly from the official update center.
- Global Git identity ensures every commit made in the lab is attributed to Meenakshi.